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California needs a million EV charging stations — but that's 'unlikely' and 'unrealistic'

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California will have to build public charging stations at an unprecedented — and some experts say unrealistic — pace to meet the needs of the 7 million electric cars expected on its roads in less than seven years.

The sheer scale of the buildout has alarmed many experts and lawmakers, who fear that the state won't be prepared as Californians purchase more electric cars.

A million public chargers are needed in California by the end of 2030, according to the — almost 10 times more than the in December. To meet that target, 129,000 new stations — more than seven times the current pace — must be built every year for the next seven years. Then the pace would have to accelerate again to reach a target of 2.1 million chargers in 2035.

Local progress toward chargers

Charger network needed to meet state EV mandate

A robust network of public chargers — akin to the state's more than — is essential to ensure that drivers will have the confidence to purchase electric vehicles over the next several years.

"It is very unlikely that we will hit our goals, and to be completely frank, the EV goals are a noble aspiration, but unrealistic," said Stanford professor Bruce Cain, who co-authored a detailing California's electric vehicle charging problems. "This is a wakeup call that we address potential institutional and policy obstacles more seriously before we commit blindly."

Under California's landmark , a pillar of Gov. Gavin Newsom's , 68% of all new 2030 model cars sold in the state must be zero emissions, increasing to 100% for 2035, when 15 million electric cars are expected in California.

"We're going to look really silly if we are telling people that they can only buy electric vehicles, and we don't have the charging infrastructure to support that," said Assemblymember Jesse Gabriel, a Democrat from Encino a package of unsuccessful bills last year aimed at expanding access to car chargers.

"We are way behind where we need to be," Gabriel told CalMatters.

Big obstacles stand in the way of amping up the pace of new charging stations in public places. California will need billions of dollars in state, federal and private investments, streamlined city and county permitting processes, and accelerated efforts by utilities to connect chargers to the grid.

State officials also are tasked with ensuring that charging stations are available statewide, in where private companies are reluctant to invest, and that they are reliable and functioning whenever drivers pull up.

In Pacific Gas & Electric's vast service area, home to 40% of all Californians, electric car purchases are moving twice as fast as the buildout of charging stations, said Lydia Krefta, the utility's director of clean energy transportation. Californians now own battery-powered cars.

Patty Monahan, who's on the Energy Commission, the state agency responsible for funding and guiding the ramp-up, told CalMatters that she is confident that California can build the chargers its residents need in time.

The agency's estimate of the current chargers is likely an undercount, she said. In addition, fast-charging stations could play a bigger role than initially projected, meaning hundreds of thousands of fewer chargers might be needed. Also, as the ranges and charging speeds on cars improve, there may be less demand for public chargers.

"California has a history of defying the odds," Monahan said. "We have a history of advancing clean cars, clean energy, writ-large. We have naysayers left and right saying you can't do it, and then we do it."

Barriers to private investments: an uncertain market

On a September day last year, Monahan spoke behind a podium in the parking lot of a Bay Area grocery store. A row of newly constructed car chargers rose behind her.

"Let's celebrate for a moment," she said.

California had met its goal of 10,000 fast electric chargers statewide — two years ahead of a set in 2018.

Fast chargers like the new ones at the grocery store are increasingly seen as critical to meeting the needs of drivers. They can in 20 minutes to an hour, while the typical charger in use today, a slower Level 2, takes from four to 10 hours.

But installing and operating fast chargers is an expensive business — one that doesn't easily turn a profit.

can cost up to $117,000, according to a 2023 study. And in California, it could be even more — between $122,000 and $440,000 each, according to , although the Energy Commission said the range was $110,000 to $125,000 for .

Most of America's publicly traded charger companies have been forced to seek more financing, lay off workers and slow their network build outs, analysts said. , for instance, has seen its share price crater, as has , which specializes in selling the slower, Level 2 hardware.

California stands apart from other states — it has by far the most chargers and electric car sales, and more incentives and policies encouraging them.

Tesla, America's top-selling electric car manufacturer, dominates fast-charging in both California and the U.S. — but the company didn't get into the business to sell charges to drivers; it got into the charger business to sell its electric cars. Initially Tesla Superchargers were exclusive to its drivers, but other EV drivers can use them after Tesla and other automakers.

Tesla's manufacturing prowess, supply chain dominance and decade-plus of experience with fast chargers have given it an edge over competitors — a coterie of unprofitable, publicly traded startups, as well as private companies that often benefit from public subsidies, according to analysts.

"All the automakers joined forces with their biggest competitor," said Loren McDonald, chief executive of the consulting firm EVAdoption. "If that doesn't tell you how bad fast-charging networks and infrastructure were, I don't know what else does."

Now Tesla is showing amid slumping car sales, and 500-member Supercharger team in April. Then chief executive Elon Musk said in May that he would to expand the network and some fired workers.

In California, Electrify America, a privately held company, was created by Volkswagen as a settlement for for its gas-powered cars. The company is spending $800 million on California chargers, building a robust network of 260 stations, with more than half in low-income communities, including , Imperial County.

The problem is Electrify America was ranked dead last in a last year, and its chargers have been plagued by reliability problems and customer complaints. The California Air Resources Board in January to "strive to achieve

charger reliability consistent with the state of the industry." A company spokesperson said the dissatisfaction showed "an industry in its growth trajectory." There are signs of improvement, based on of this year.

Startups continue to jump into the charging business, with the number of companies offering fast chargers growing from 14 in 2020 to 41 in 2024, EVAdoption said. Seven carmakers formed a to build a 30,000-charger network in North America. And such as Circle K are offering more charging because electric car customers spend more time shopping while waiting for their rides to juice up.

But the realization that charging is a costly business has set in on Wall Street, and that doesn't seem likely to change anytime soon. "Can public EV fast-charging stations be profitable in the United States?" the .

"The fervor, the excitement from the investor base, has definitely dwindled quite a bit, given the prospects that EV adoption in the U.S. is going to be slower, revenue growth is really slower, the path to profitability is going to be slower, and they might need more capital than everyone originally expected," said Christopher Dendrinos, a financial analyst who covers electric car charging companies for the investment bank RBC Capital Markets.

The stakes are high for California when it comes to encouraging investments in expensive fast chargers: If 63,000 additional ones were built, California might need 402,000 fewer slower Level 2 chargers in 2030, according to an alternative by the Energy Commission.

Billions of public dollars: Will it be enough?

Nationwide $53 billion to $127 billion in private investments and public funding is needed by 2030 to build chargers for about 33 million electric cars, . Of that, about half would be for public chargers.

Congress and the Biden administration have set aside $5 billion for a national network of fast chargers. So far only 33 in eight locations have been built, but more than 14,000 others are in the works, according to the Federal Highway Administration. California's share of the federal money totals $384 million; about 500 fast chargers will be built with an initial $40.5 million, said Energy Commission spokesperson Lindsay Buckley.

In addition, the state has spent $584 million to build more than through its Clean Transportation Program, funded by fees drivers pay when they register cars. The Legislature for an additional decade last year.

Newsom has committed to spending $1 billion through 2028 on chargers with his "," Buckley said. But this year Newsom and the Legislature trimmed $167 million from the charger budget as the state faces a record deficit. A lobbyist for the Electric Vehicle Charging Association said "the state pullback sends a very challenging message" to the industry.

California's commitment to charger funding is "solid," despite the cuts, Buckley said. They have not yet estimated the total investment needed in California to meet the targets.

But Ted Lamm, a UC Berkeley Law researcher who studies electric car infrastructure, said the magnitude of building what California needs in coming years likely dwarfs the public funding available.

State and federal programs will "only fund a fraction," and the state needs to spend that money on lower-income communities, he said.

Another possible funding source is California's . The program requires carbon-intensive fuel companies to pay for cleaner-burning transportation. Utilities get credits and use that money to pay for chargers, rebates to car buyers and grid improvements, said Laura Renger, executive director of the California Electric Transportation Coalition, which represents utilities.

"I think with that, we would have enough money," Renger said. She said the program's overhaul could help utilities invest "billions" in chargers and other electric car programs over the next two decades.

Backlogged local permits and grid delays

One of the biggest barriers to more chargers isn't money. It's that cities and counties are slow to approve plans for the vast number of stations needed.

State officials only have so much political power to compel local jurisdictions to do what they want — a reality made by the , for instance. California relies on grants and persuasion to accomplish its goals, and the slow buildout of chargers shows how those strategies can fall short, said Stanford's Cain.

"The locals cannot be compelled by regulatory agencies to make land and resources available for what the state wants to achieve," Cain said.

The same obstacles have marked the state's broader effort to electrify California and switch to clean energy. Local opposition and environmental reviews sometimes hold up and for years.

California has created a "culture of regulation that emphasizes the need to be extra careful and extra perfect, but this takes an incredible amount of time," Steve Bohlen, senior director of government affairs at Lawrence Livermore National Laboratory, said last month at the .

"We're moving into a period of rapid change, and so perfect can't be the enemy of the good."

Chargers aren't as complicated as large-scale solar or . But most chargers installed in public spaces do need a land-use or encroachment permit, among other approvals. California has requiring local jurisdictions to streamline permits for chargers. What's more, the Governor's Office of Business Development now grades cities and counties using a and displaying who has, or hasn't, made life easier for car charger builders. But these strategies only go so far.

"It doesn't matter how many requirements you put on (local governments)," Lamm said. "If they just don't have the time in the day to do it … it's going to sit in the backlog, because that's how it works."

The delays have consequences. Getting a station permitted in California, on average, than the national average, Electrify America reported. Designing and constructing a station in California can cost on average than in other states because of delays in permitting and grid connections. A utility on average takes 17 weeks after work is completed to connect chargers to the grid, Electric America said.

Powering large charging projects often requires grid upgrades, which can take a year or more for approval, said Chanel Parson, a director at Southern California Edison. Supply chain issues also make getting the right equipment a challenge.

Edison, which has a 10-year plan to meet expected demand, has asked the utilities commission for approval to upgrade the grid where it anticipates high charging demand.

"Every EV charging infrastructure project is a major construction project," Parson said. "There are a number of variables that influence how long it takes to complete the project."

Impatient with broken chargers, bad service

Inspired to help the nation reduce its dependence on fossil fuels, Zach Schiff-Abrams of Los Angeles bought a Genesis GV60. As a renter, he has relied on public charging, primarily using Electrify America stations — and that's been his biggest problem about owning an electric car.

Charging speeds have been inconsistent, he said, with half-hour sessions providing only a 15 to 30% charge, and he often encounters broken chargers.

"I believe in electrical, so I'm really actually trying to be a responsible consumer," Schiff-Abrams said. "I want to report them when they're down, but the customer service is horrible."

For years, the reliability of charging networks has been a well-documented problem. Only in a 2022 study. The growth of the EV market has put increasing strain on public charging stations, .

In , the California Air Resources Board approved a final $200 million spending plan for Electrify America — but not before board chair Liane Randolph scolded its CEO.

Randolph — arguably one of America's top climate regulators — told CEO Robert Barrosa about an exchange she had with his company's customer service line after finding a broken charger at a station along Interstate-5.

"It didn't work," Randolph said during the board meeting. "Called the customer service line, waited like 10-ish minutes. ...(The charger) was showing operable on the app and the guy goes, 'oh, my data is showing me that it has not had a successful charge in three days.'"

"These issues are not easy," Barrosa responded. "Our head is not in the sand," he earlier. "We are listening to customers."

But Randolph, addressing journalists at a , pushed back against the idea that because the transition to electric vehicles is happening gradually that it's a failure. Many people will rely on charging at home or work, and batteries are becoming more efficient.

"The infrastructure is continuing to be rolled out at a rapid pace," Randolph said. "It doesn't all have to be perfect instantly. It's a process. And it's a process that's continuing to move."

Data journalists Erica Yee and Arfa Momin contributed to this report. North Bay Business Journal compiled local data.

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